1.36 draw a simple diagram to represent the positions of the ions in a crystal of sodium chloride

g) Covalent substances

Students will be assessed on their ability to:

1.37 describe the formation of a covalent bond by the sharing of a pair of electrons between two atoms

1.38 understand covalent bonding as a strong attraction between the bonding pair of electrons and the nuclei of the atoms involved in the bond

1.39 explain, using dot and cross diagrams, the formation of covalent compounds by electron sharing for the following substances:

- i hydrogen
- ii chlorine
- iii hydrogen chloride
- iv water
- v methane
- vi ammonia
- vii oxygen
- viii nitrogen
- ix carbon dioxide
- x ethane
- xi ethene

1.40 recall that substances with simple molecular structures are gases or liquids, or solids with low melting points

1.41 explain why substances with simple molecular structures have low melting points in terms of the relatively weak forces between the molecules

1.42 explain the high melting points of substances with giant covalent structures in terms of the breaking of many strong covalent bonds

1.43 draw simple diagrams representing the positions of the atoms in diamond and graphite

1.44 explain how the uses of diamond and graphite depend on their structures, limited to graphite as a lubricant and diamond in cutting

h) Metallic crystals

Students will be assessed on their ability to:

1.45 describe a metal as a giant structure of positive ions surrounded by a sea of delocalised electrons

1.46 explain the malleability and electrical conductivity of a metal in terms of its structure and bonding

i) Electrolysis

Students will be assessed on their ability to:

1.47 understand an electric current as a flow of electrons or ions

1.48 understand why covalent compounds do not conduct electricity

1.49 understand why ionic compounds conduct electricity only when molten or in solution

1.50 describe simple experiments to distinguish between electrolytes and non-electrolytes

1.51 recall that electrolysis involves the formation of new substances when ionic compounds conduct electricity

1.52 describe simple experiments for the electrolysis, using inert electrodes, of molten salts such as lead(II) bromide

1.53 describe simple experiments for the electrolysis, using inert electrodes, of aqueous solutions of sodium chloride, copper(II) sulfhate and dilute sulfhuric acid and predict the products

1.54 write ionic half-equations representing the reactions at the electrodes during electrolysis

1.55 recall that one faraday represents one mole of electrons

1.56 calculate the amounts of the products of the electrolysis of molten salts and aqueous solutions

Section 2: Chemistry of the elements

a) The Periodic Table

b) The Group 1 elements – lithium, sodium and potassium

c) The Group 7 elements – chlorine, bromine and iodine

d) Oxygen and oxides

e) Hydrogen and water

f) Reactivity series

g) Tests for ions and gases

a) The Periodic Table

Students will be assessed on their ability to:

2.1 understand the terms group and period

2.2 recall the positions of metals and non-metals in the Periodic Table

2.3 explain the classification of elements as metals or non-metals on the basis of their electrical conductivity and the acid-base character of their oxides

2.4 understand why elements in the same group of the Periodic Table have similar chemical properties

2.5 recall the noble gases (Group 0) as a family of inert gases and explain their lack of reactivity in terms of their electronic configurations

b) The Group 1 elements – lithium, sodium and potassium

Students will be assessed on their ability to:

2.6 describe the reactions of these elements with water and understand that the reactions provide a basis for their recognition as a family of elements

2.7 recall the relative reactivities of the elements in Group 1

2.8 explain the relative reactivities of the elements in Group 1 in terms of distance between the outer electrons and the nucleus

c) The Group 7 elements – chlorine, bromine and iodine

Students will be assessed on their ability to:

2.9 recall the colours and physical states of the elements at room temperature

2.10 make predictions about the properties of other halogens in this group

2.11 understand the difference between hydrogen chloride gas and hydrochloric acid

2.12 explain, in terms of dissociation, why hydrogen chloride is acidic in water but not in methylbenzene

2.13 recall the relative reactivities of the elements in Group 7

2.14 describe experiments to show that a more reactive halogen will displace a less reactive halogen from a solution of one of its salts

2.15 understand these displacement reactions as redox reactions

d) Oxygen and oxides

Students will be assessed on their ability to:

2.16 recall the gases present in air and their approximate percentage by volume

2.17 describe how experiments involving the reactions of elements such as copper, iron and phosphorus with air can be used to determine the percentage by volume of oxygen in air

2.18 describe the laboratory preparation of oxygen from hydrogen peroxide

2.19 describe the reactions with oxygen in air of magnesium, carbon and sulphur, and the acid-base character of the oxides produced

2.20 describe the laboratory preparation of carbon dioxide from calcium carbonate and dilute hydrochloric acid

2.21 describe the formation of carbon dioxide from the thermal decomposition of metal carbonates such as copper(II) carbonate

2.22 recall the properties of carbon dioxide, limited to its solubility and density

2.23 explain the use of carbon dioxide in carbonating drinks and in fire extinguishers, in terms of its solubility and density

2.24 recall the reactions of carbon dioxide and sulphur dioxide with water to produce acidic solutions

2.25 recall that sulphur dioxide and nitrogen oxides are pollutant gases which contribute to acid rain, and describe the problems caused by acid rain

e) Hydrogen and water

Students will be assessed on their ability to:

2.26 describe the reactions of dilute hydrochloric and dilute sulfhuric acids with magnesium, aluminium, zinc and iron

2.27 describe the combustion of hydrogen

2.28 describe the use of anhydrous copper(II) sulfate in the chemical test for water

2.29 describe a physical test to show whether water is pure

f) Reactivity series

Students will be assessed on their ability to:

2.30 recall that metals can be arranged in a reactivity series based on the reactions of the metals and their compounds: potassium, sodium, lithium, calcium, magnesium, aluminium, zinc, iron, copper, silver and gold

2.31 describe how reactions with water and dilute acids can be used to deduce the following order of reactivity: potassium, sodium, lithium, calcium, magnesium, zinc, iron, and copper

2.32 deduce the position of a metal within the reactivity series using displacement reactions between metals and their oxides, and between metals and their salts in aqueous solutions

2.33 understand oxidation and reduction as the addition and removal of oxygen respectively

2.34 understand the terms: redox, oxidising agent and reducing agent

2.35 recall the conditions under which iron rusts

2.36 describe how the rusting of iron may be prevented by grease, oil, paint, plastic and galvanising

2.37 understand the sacrificial protection of iron in terms of the reactivity series

g) Tests for ions and gases

Students will be assessed on their ability to:

2.38 describe simple tests for the cations

- i Li^+, Na^+, K^+, Ca^{2+} using flame tests
- ii NH_4^+ using sodium hydroxide solution and identifying the ammonia evolved
- iii Cu^{2+}, Fe^{2+} and Fe^{3+} using sodium hydroxide solution

2.39 describe simple tests for the anions:

- i $Cl^–$, $Br^–$ and $I^–$, using dilute nitric acid and silver nitrate solution
- ii $SO_4^{2–}$, using dilute hydrochloric acid and barium chloride solution
- iii $CO_3^{2–}$, using dilute hydrochloric acid and identifying the carbon dioxide evolved

2.40 describe simple tests for the gases

- i hydrogen
- ii oxygen
- iii carbon dioxide
- iv ammonia
- v chlorine